

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY



Department of Computer Science and Engineering

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CSE2104 Offline-2

You are given a list of books, each with a title, author and publication year. Your task is to sort the books by the publication year in **descending order** (most recent first) and if two books have the same publication year, sort them lexicographically by author in **ascending order**. If both the publication year and author are the same, sort the books lexicographically by title in **ascending order**. Implement this sorting using the **Selection Sort** algorithm. The input consists of an integer N (the number of books), followed by N lines of book details, where each line contains the title, author and publication year. Output the books in the sorted order, each on a new line in the format: "Title, Author, Publication Year".

Input Format

- The first line contains an integer N, the number of books ($1 \leq N \leq 1000$).
- Each of the next N lines contains a book's information in the format:
 - Title, Author, Publication Year
 - The Title is a string ($1 \leq \text{length of title} \leq 100$)
 - The Author is a string ($1 \leq \text{length of author name} \leq 100$)
 - The Publication Year is an integer ($1900 \leq \text{year} \leq 2025$)

Output Format

Print the books in sorted order based on the criteria above, one book per line in the following format:

- Title, Author, Publication Year

Algorithm

1. Read the integer N (number of books).
2. For each of the N books, read a line and parse the Title, Author, and Publication Year separated by commas.
3. Store each book as a struct with fields: bName (title), author, and y (publication year).
4. Apply **Selection Sort** on the list of books using the following multi-key comparison:
 - Primary key: Publication Year in **descending** order (larger year comes first).
 - Secondary key: Author name in **ascending** lexicographic order (if years are equal).
 - Tertiary key: Book title in **ascending** lexicographic order (if year and author are equal).
5. In each pass of Selection Sort, find the maximum element (per the above criteria) from the unsorted portion and swap it to its correct position.
6. After sorting, print each book on a new line in the format: Title, Author, Publication Year.

Source Code (C++)

```
#include<bits/stdc++.h>
using namespace std;
struct Book
{
    string bName;
    string author;
    int y;
};

void selectionSort(vector<Book> &b, int n)
{
    for(int i=0;i<n-1;i++)
    {
        int max=i;
        for(int j=i+1;j<n;j++)
        {
            if(b[j].y==b[max].y)
            {
                if(b[j].author<b[max].author)
                {
                    max=j;
                }
                else if(b[j].author==b[max].author)
                {
                    if(b[j].bName<b[max].bName)
                    {
                        max=j;
                    }
                }
            }
            else if(b[j].y>b[max].y)
            {
                max=j;
            }
        }
        swap(b[i],b[max]);
    }
}

int main()
{
    int n;
    cin>>n;
    cin.ignore();
    vector<Book> b(n);
    for(int i=0;i<n;i++)
    {
        string line;
        getline(cin, line);

        stringstream ss(line);
```

```

getline(ss, b[i].bName, ',');
getline(ss, b[i].author, ',');

if (b[i].author[0] == ' '){
    b[i].author.erase(0, 1);
}
ss >> b[i].y;

}
selectionSort(b,n);
for(int i=0;i<n;i++)
{
    cout << b[i].bName << ", "
        << b[i].author << ", "
        << b[i].y << endl;

}
}

```

Sample Input	Sample Output
5 Sapiens, Yuval Noah Harari, 2011 21 Lessons for the 21st Century, Yuval Noah Harari, 2018 Educated, Tara Westover, 2018 Becoming, Michelle Obama, 2018 Money: Vintage Minis, Yuval Noah Harari, 2018	Becoming, Michelle Obama, 2018 Educated, Tara Westover, 2018 21 Lessons for the 21st Century, Yuval Noah Harari, 2018 Money: Vintage Minis, Yuval Noah Harari, 2018 Sapiens, Yuval Noah Harari, 2011